Meet-in-the-Middle Attacks on Full ChiLow

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[2]COSIC, KU Leuven, Leuven, Belgium

[3]Nanyang Technological University, Singapore

ChiLow

- Tweakable block cipher

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- Published at Eurocrypt 2025
- Designed by Yanis Belkheyar, Patrick Derbez, Shibam Ghosh, Gregor Leander, Silvia Mella, Léo Perrin, Shahram Rasoolzadeh, Lukas Stennes, Siwei Sun, Gilles Van Assche and Damian Vizár

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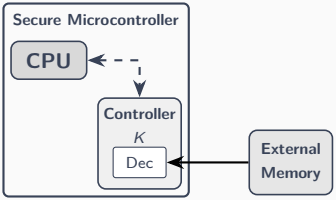

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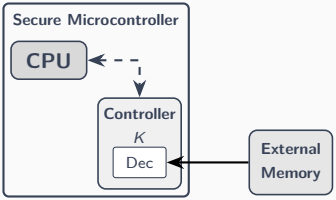
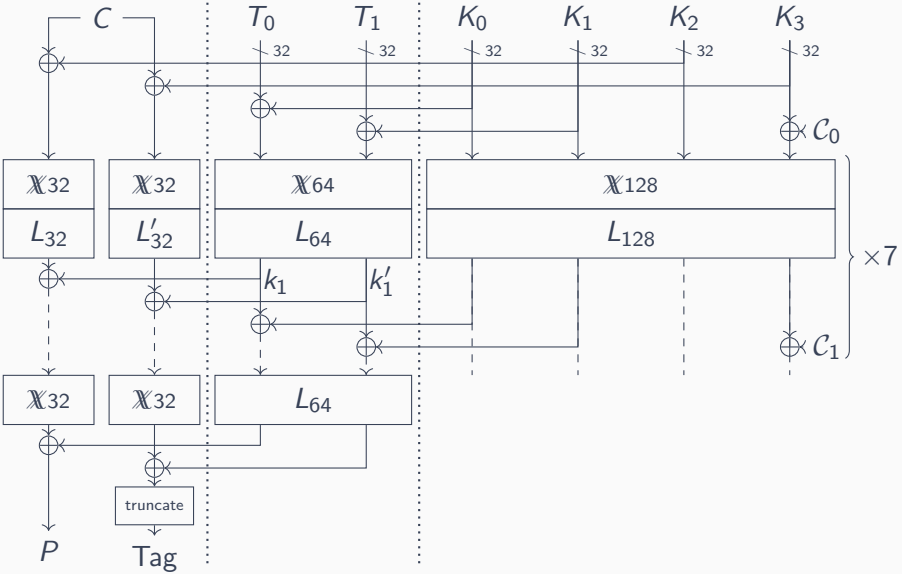


ChiLow-32


C
T0
T1
K0
K1
K2
K3
32
C0
X32
X32
X64
X128
L32
L'32
L64
L128
×7
k1
k'1
C1
X32
X32
L64
truncate
P
Tag

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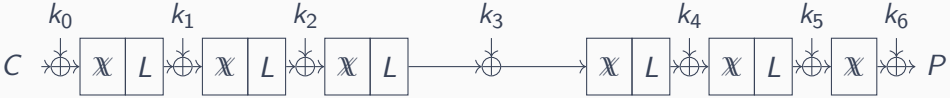

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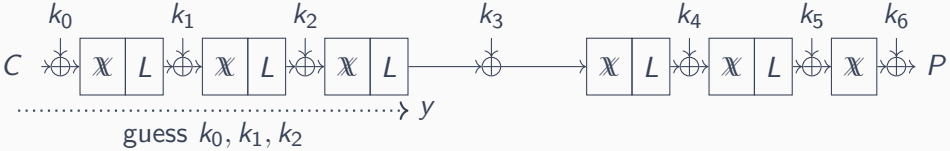

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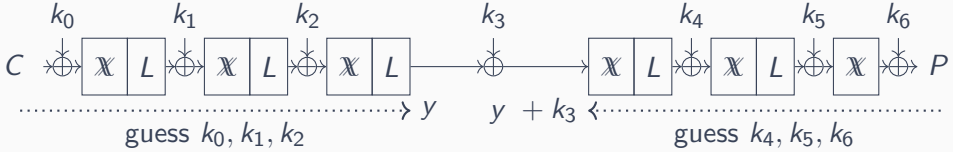

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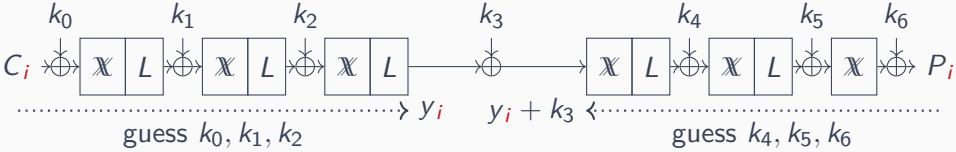

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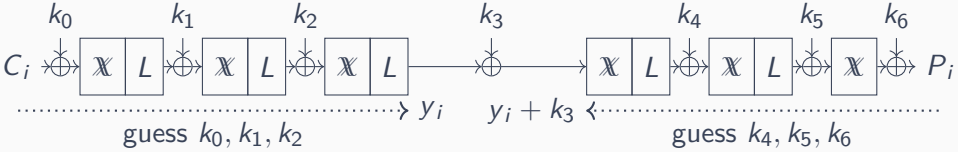

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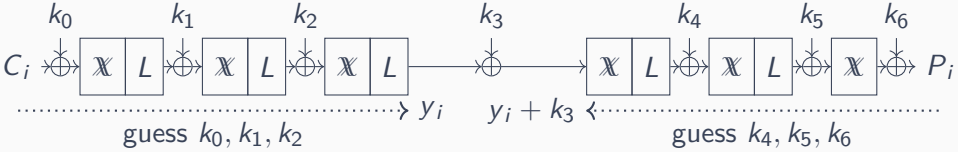

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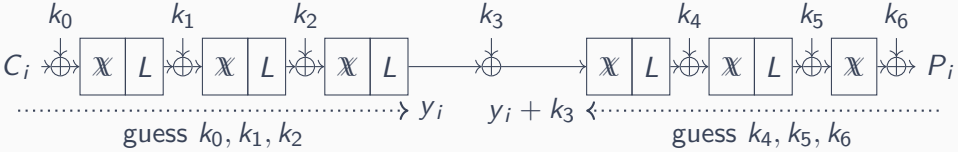

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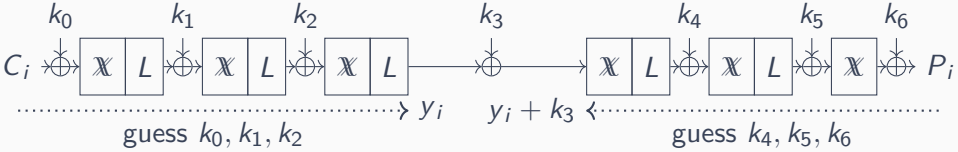

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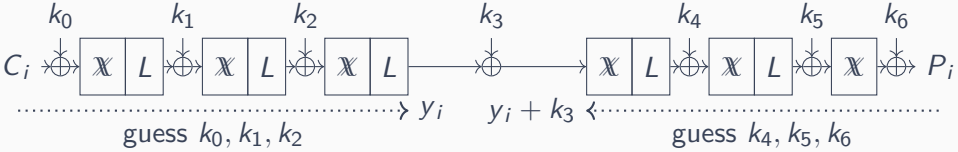

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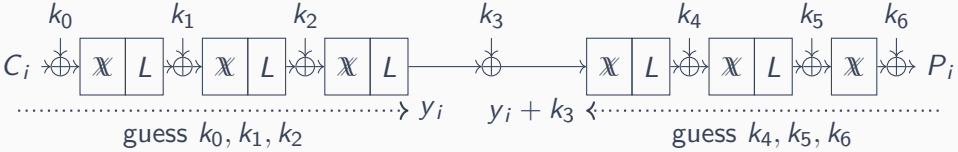

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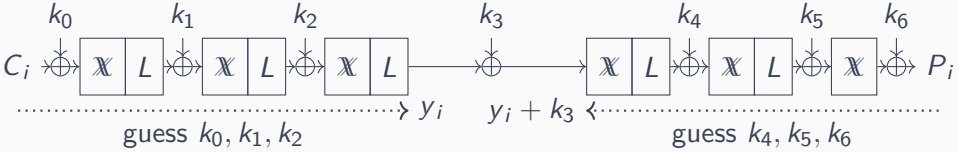

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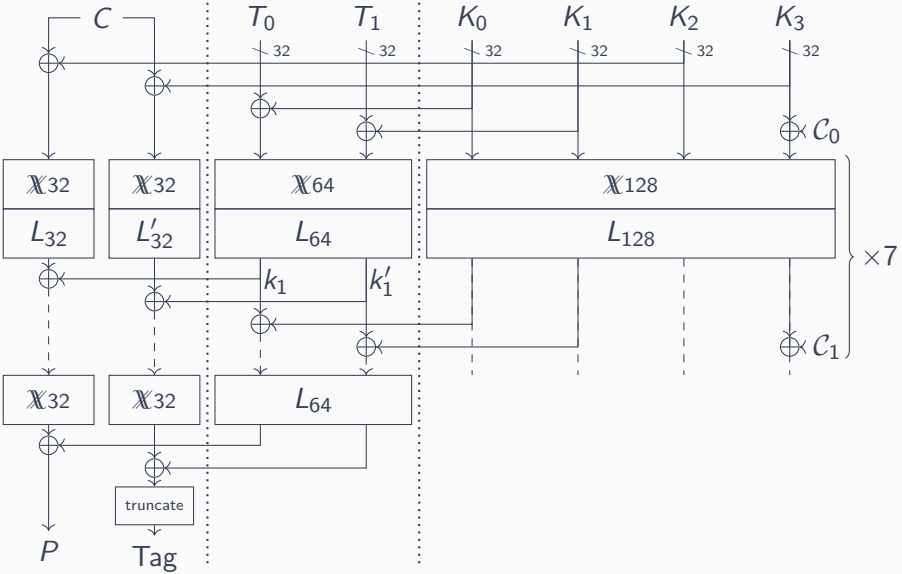

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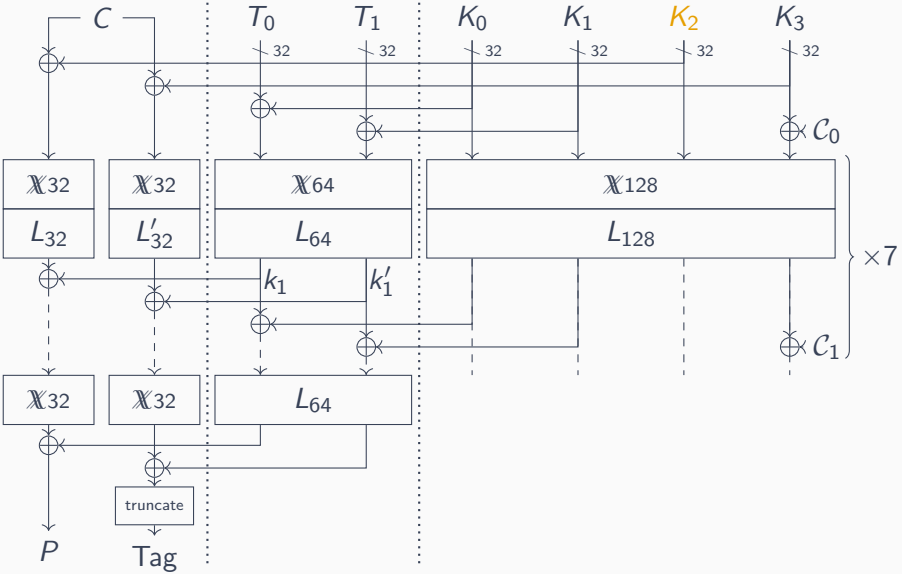

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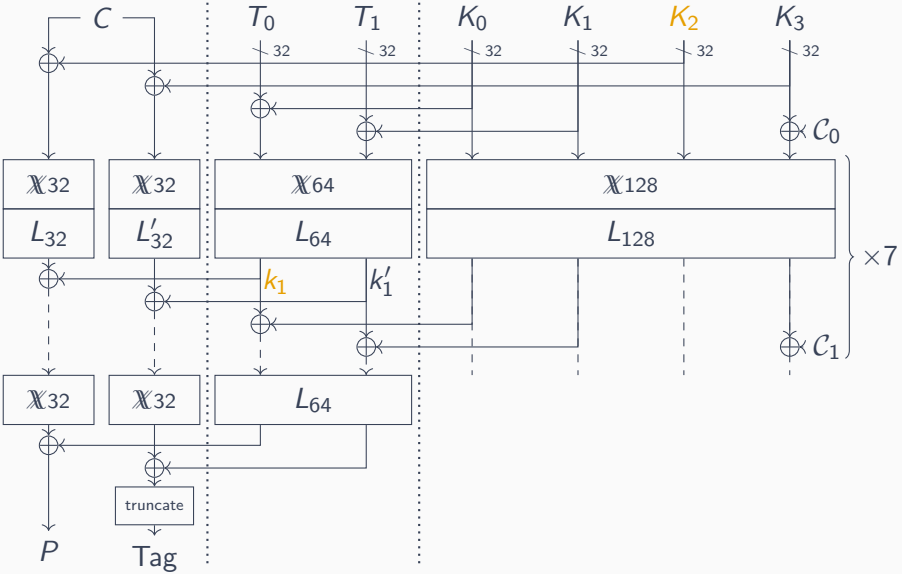

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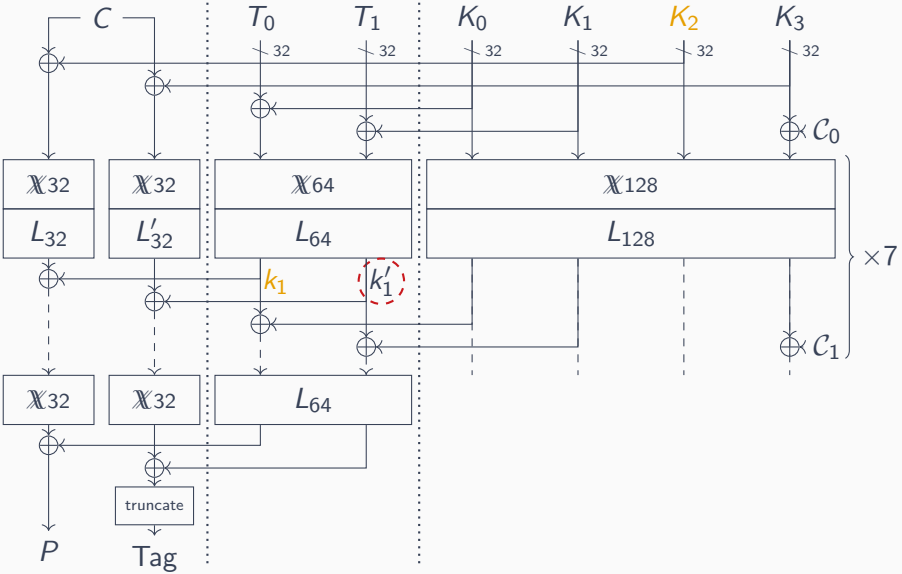

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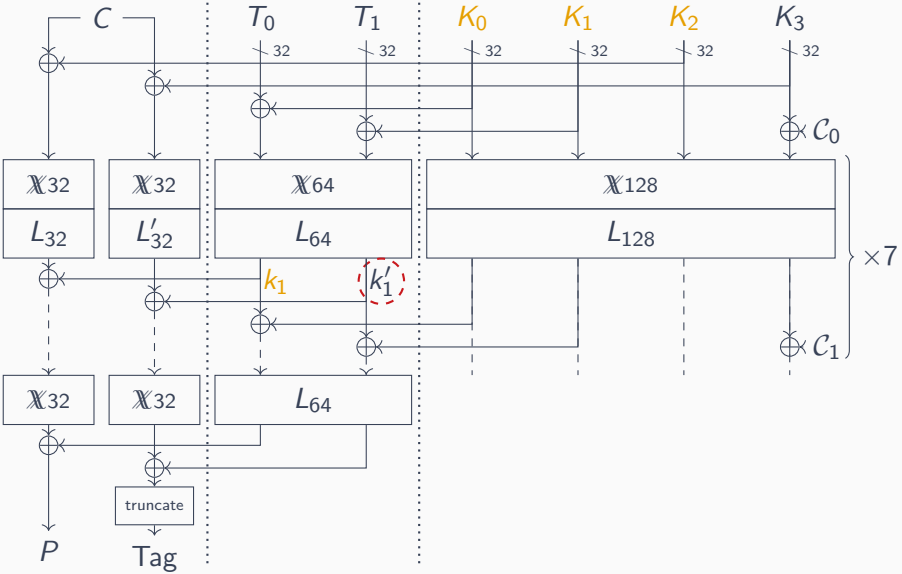

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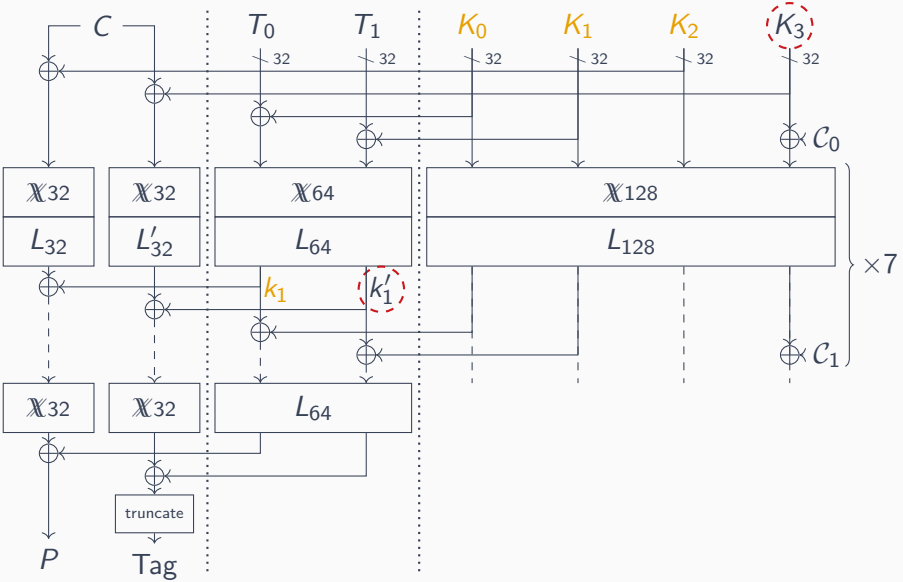

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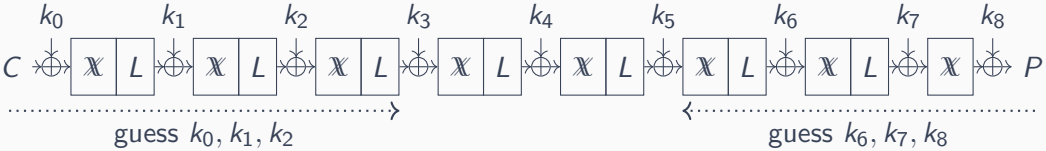

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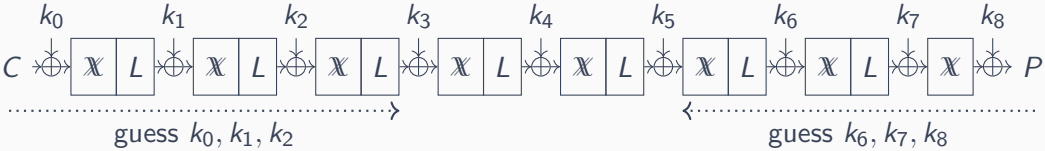

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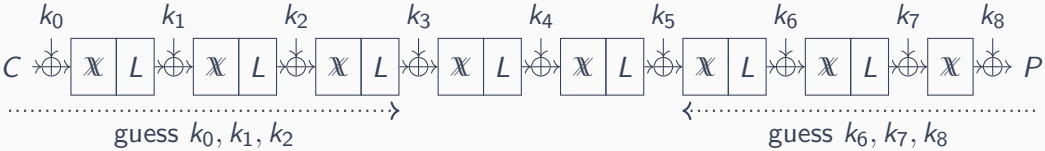

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- How much key material of k_3, k_4, k_5 do we need to know to match on one bit?

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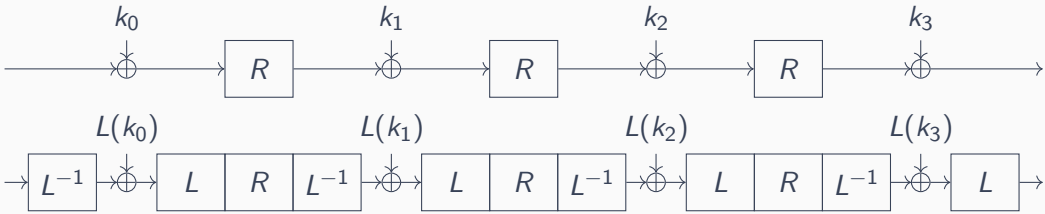

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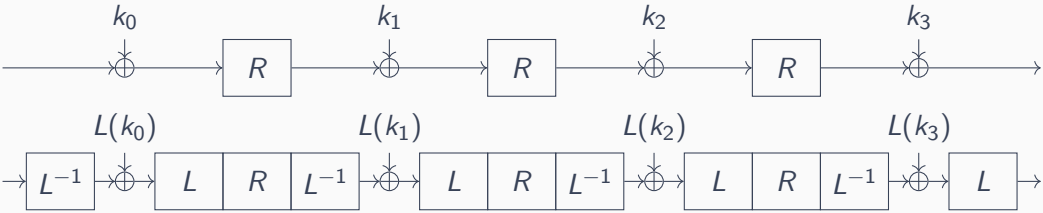

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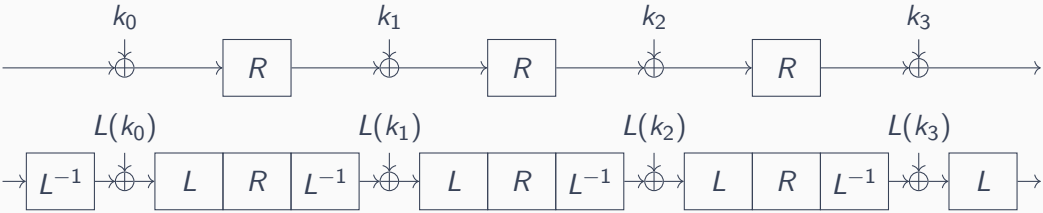

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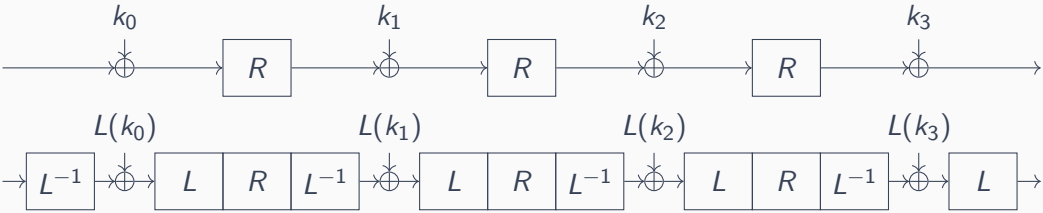

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- Represent bits of $x \in \mathbb{F}_2^n$ as $v^\mathsf{T} x = \sum_i v_i x_i$ for mask $v \in \mathbb{F}_2^n$

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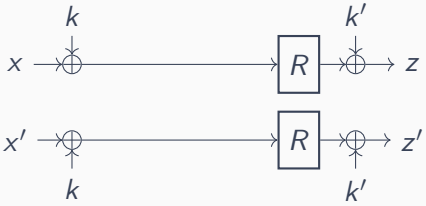

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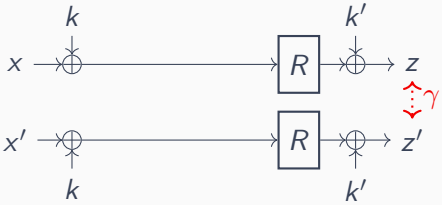

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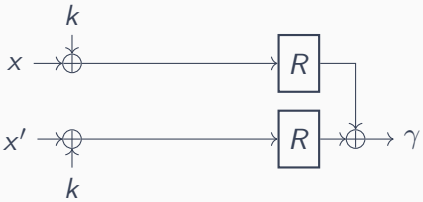

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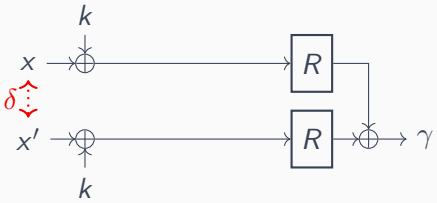

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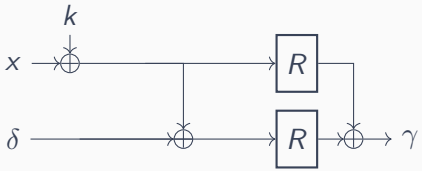

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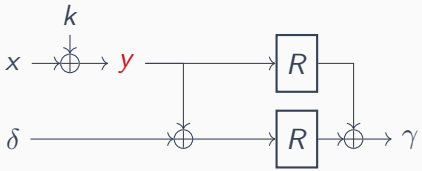

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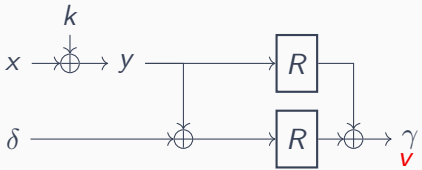

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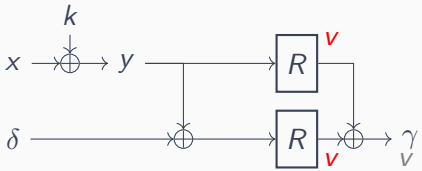

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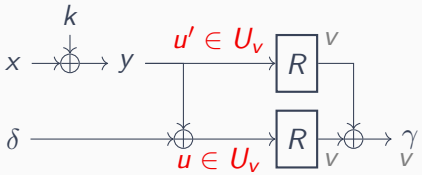

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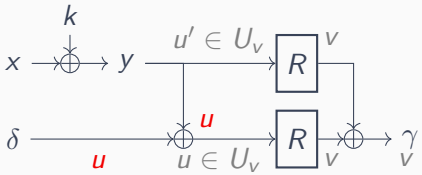

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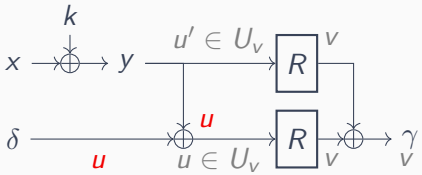

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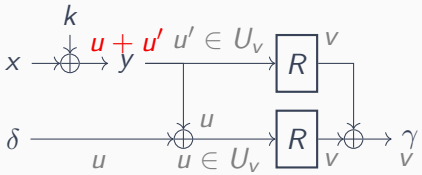

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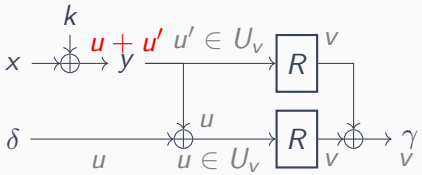

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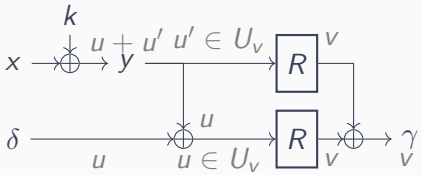

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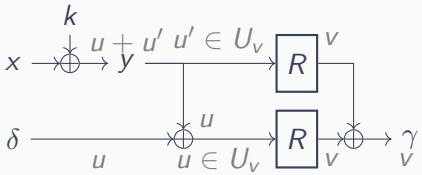

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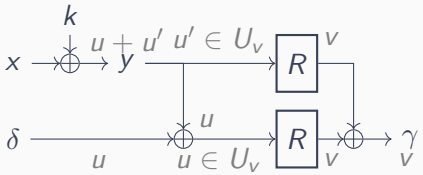

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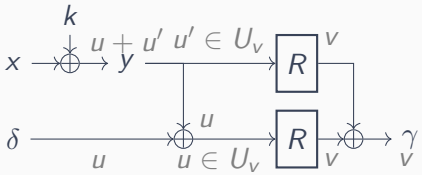

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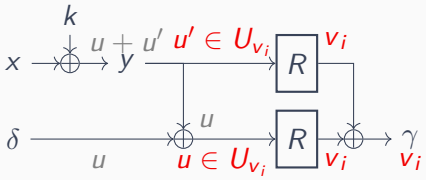

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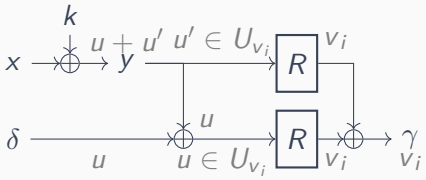

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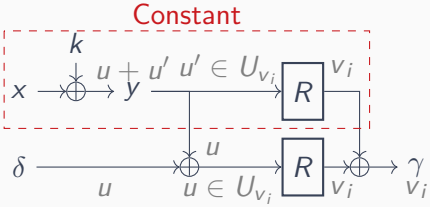

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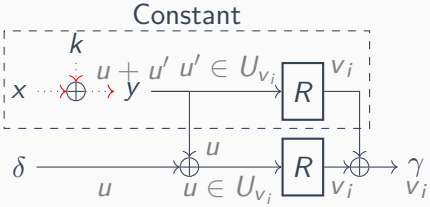

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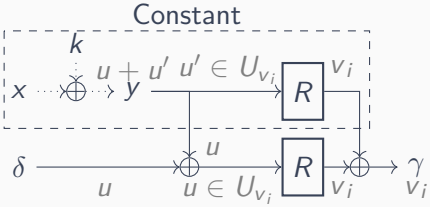

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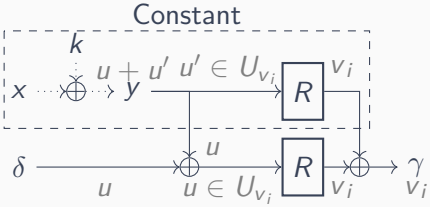

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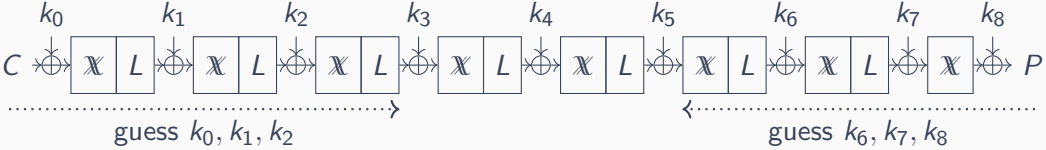
- Guess state y instead of key k
- Can be iterated over multiple rounds
- Guessing y reduces dependencies when restricting δ

State Dependencies for 2-Round ChiLow

Block Size	Number of Matching Bits	State Dependencies (/Bits)
32	1	17
	2	23
	3	27
40	1	17
	2	24
	3	28

A Known Plaintext Attack on Full ChiLow-32



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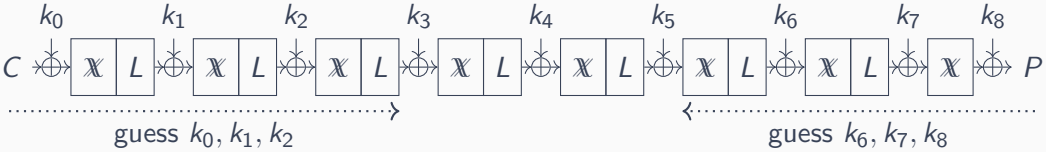

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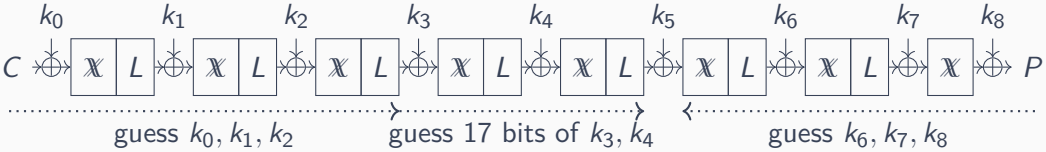

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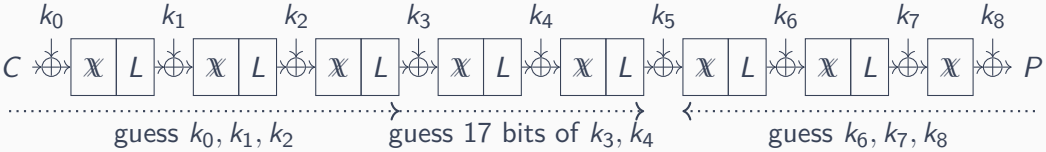

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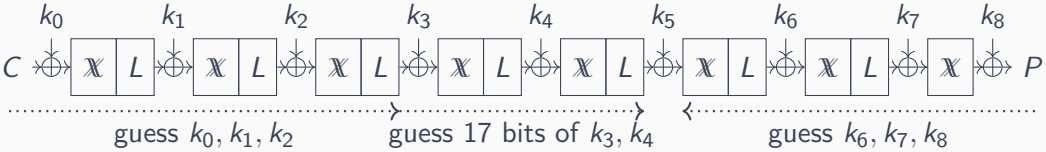

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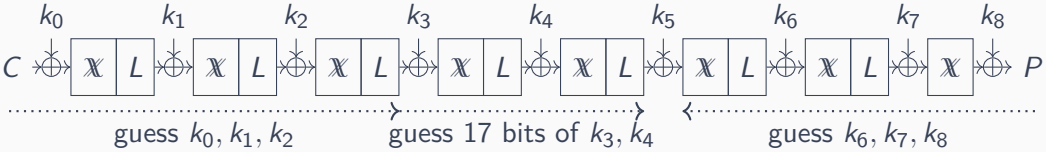

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- Complexities: $2^{120.34}$ (time), $2^{98.32}$ (memory) and 160 (data)

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- Example: $v^\mathsf{T} R(x) = x_0 + x_1 x_2 + x_2$

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- Propagate truncated differences Δ, modeled as affine spaces
- Enough to consider basis $\delta_0, \ldots, \delta_{b-1}$ and offset α of Δ

State Dependencies for Computing the Output Difference of 2-Round ChiLow

Block Size	Dimension	Non-constant Bits	State Dependency (/Bits)
32	6	2 – 7	33
	7	2 – 8	35
	8	2 – 9	37
40	6	10 – 15	40
	7	10 – 16	43
	8	10 – 17	45

An Optimized Chosen Ciphertext Attack on Full ChiLow-32

- Use 6-dimensional space (requiring 33 state bits over 2 rounds)

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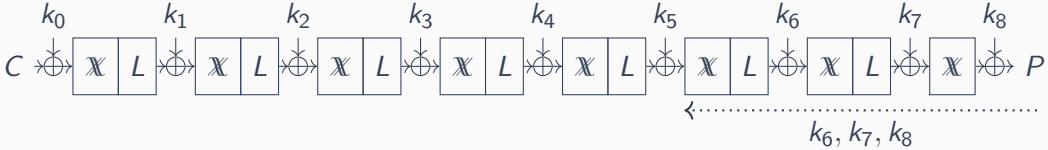

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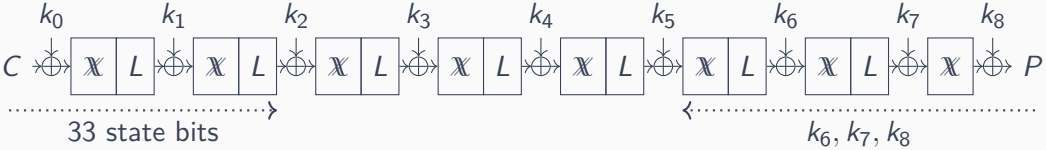

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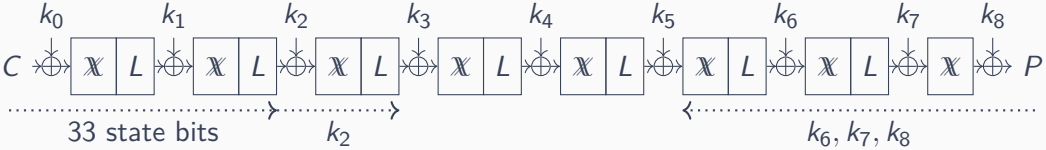

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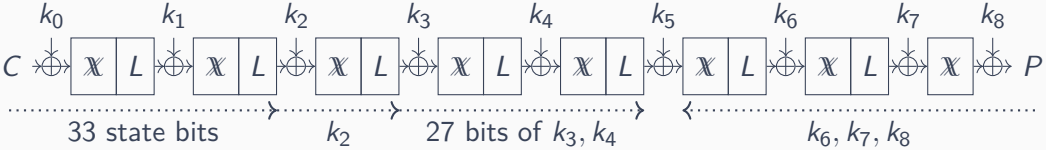

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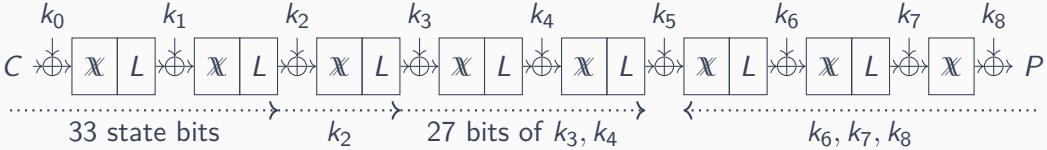

- Guess $33 + 32 + 27 = 92$ bits from the ciphertext side

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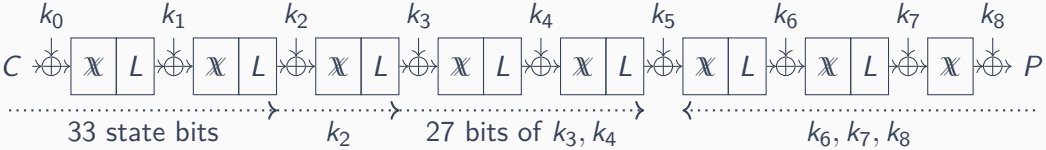

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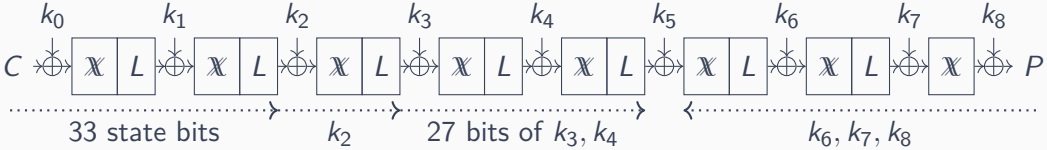

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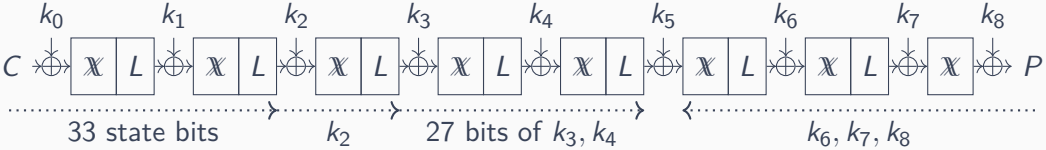

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Overview of Results

Variant	Rounds	Time	Memory	Data	Type	Ref.
ChiLow-32	4	2^{126}	–	–	MitM	[BDG+25]
	5	2^{32}	$2^{28.65}$	$2^{18.58}$ CC	Cube	[AHH+26]
	6	2^{34}	$2^{49.28}$	$2^{33.58}$ CC	Cube	[AHH+16]
	7	$2^{121.75}$	–	80 CC	Integral	[AHH+16]
	8	$2^{120.34}$	$2^{98.32}$	160 KP	MitM	This Work
	8	$2^{102.09}$	$2^{94.56}$	64 CC	MitM	This Work
	9	2^{118}	$2^{106.68}$	256 CC	MitM	This Work
ChiLow-40	4	2^{126}	–	–	MitM	[BDG+25]
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Overview of Results

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	7	$2^{121.75}$	–	80 CC	Integral	[AHH+16]
	8	$2^{120.34}$	$2^{98.32}$	160 KP	MitM	This Work
	8	$2^{102.09}$	$2^{94.56}$	64 CC	MitM	This Work
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Thank you for your attention!